Mark Jarzynski

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SUMMARY

- **Experienced Graphics Engineer** – Experienced graphics engineer with over 15 years of software development experience with a focus on graphics, content pipeline, and optimization.
- **Published Graphics Researcher** – Published three papers in major publications, including a highly referenced paper on hash function quality/speed trade-off evaluation that introduced a new class of performant hash functions tuned for multidimensional values.
- **Director & Manager** – Managed and mentored a wide range of research teams over 20+ projects, including graphics optimization techniques and a geospatial visualizer for Unreal 4.
- **Excellent Cross-Discipline Communicator** – Demonstrated ability to maintain positive and productive communication and collaboration with both partners and other disciplines.

SKILLS

Languages: C • C++ • HLSL • GLSL • C# • Python • Java • PHP • HTML • CSS • JavaScript
Graphics APIs: DirectX 11/12 • Vulkan • OpenGL
Graphics Debuggers: PIX • Nvidia Nsight Graphics/Aftermath • AMD Radeon GPU Profiler
Game Engines: Unreal Engine • Unity • Nitrous • Iron Engine
Platforms: Windows • Linux • macOS • Android • iOS • Oculus • SteamVR
Databases: MariaDB • MySQL • MSSQL • Postgres • Oracle
Source Control: Perforce • Git • Subversion • CVS • Mercurial
Tools: Visual Studio • XCode • CMake • Makefiles • GCC • LLVM • ImGui • OptiTrack • NGINX • Apache • Docker • MATLAB • JIRA • Confluence • Microsoft Suite • Google Workspace

EXPERIENCE

Graphics Engineer – Oxide Games 2021 - Present

- Extended and maintained the Decoupled Shading Engine of Nitrous, Oxide's custom game engine used for Ara History Untold and Ashes of the Singularity II.
- Developed and maintained Nitrous' graphics features and shader, including terrain, water, point lights, area lights, effects, particles, trails, and a hybrid BRDF model (GGX + OrenNayar).
- Worked closely with design and art departments to achieve the desired VFX, lighting, and game visuals and to ensure tools were easily usable via ImGui interfaces and debug features.
- Optimized Nitrous' terrain shader using an expanded stochastic blended hex tiling approach, resulting in a flexible terrain tool with 8 layers of control, no apparent repeats, and 50% increased GPU throughput, enabling the title to operate on limited systems including the Steam Deck.
- Created dynamic world terrain modifications to create flat building foundations and smooth transitions to city infrastructure via an efficient 16-coefficient implementation of De Casteljau's algorithm that provides artistic control and preserves mountain peaks.
- Reimplemented tone mapping system with new techniques that better preserve hue in HDR and a supplementary interpolation between new and old systems for artists to select desired visuals.
- Developed a dynamic infrastructure renderer through extensive collaboration with art and design to enable real-time city generation with live debugging, cinematics support, advanced building and terrain aware pathing, and naturalizing, era, and deterioration modifications.
- Wrote SSE-optimized OBB collision to the graphics acceleration structure to efficiently enable long light beam visuals.
- Implemented an anisotropic BRDF to support shading hair, metal, and fur.
- Created deterministic pseudo random number generators that are now used studio-wide.
- Added status system to efficiently handle status state and logic across a large number of units.

Technical Director – UMBC, Imaging Research Center 2011 – 2021

- Led, supervised, and educated teams of up to 13 students and staff on a variety of projects.
- Led 20+ research projects with focuses across ray tracing, hash functions, parallel computing, procedural generation, motion capture, photogrammetry, data visualization, and XR.
- Maintained and supported a 10x10 meter VR space and 100 camera photogrammetry facility to support various research projects.
- Equipped teams with the best tools and resources available to support their projects, personal development, and future career opportunities.

Systems Administrator & Programmer – UMBC, Imaging Research Center 2011 – 2014

- Developed and maintained mission critical software to meet research needs.
- Ensured researcher needs were being met by student teams through leadership and mentorship.
- Maintained department's infrastructure, servers, and desktops.

PUBLISHED GAMES

Ashes of the Singularity II — TBA
Sins of a Solar Empire II: Paths to Power — 2025
Ara History Untold — 2024

PUBLICATIONS

"Generalized Decoupled and Object Space Shading System", ESGR — 2022
"Hash Functions for GPU Rendering", JCGT — 2020
"Extending CoNavigator into a Collaborative Digital Space", GROUP '20 — 2020

EDUCATION

Ph.D. in Computer Science; University of Maryland, Baltimore County — 2024 - Present
Dissertation: *Hashing Out Performance, Evaluating and Optimizing Hash Functions for GPU Rendering*

B.S. in Computer Science; University of Maryland, Baltimore County — 2011

AWARDS

Eagle Scout; Scouts BSA Troop 447 — 2006